EXISTENCE AND UNIQUENESS OF THE NORMAL CLOSURE OF A FINITE EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The following definition-normal-closure (source lines 2340-2345) names the field constructed here; the inside-a-given-normal-extension variant is lemma-normal-closure-inside-normal later in the chapter.

Editorial note. The label lemma-normal-closure names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-splitting-field (smallest splitting extension exists, is normal and unique up to isomorphism, source lines 2265-2293), lemma-map-into-algebraic-closure (embeddings into an algebraic closure extend along algebraic extensions), lemma-intersect-normal (the intersection of two normal subextensions in a common overfield is normal), lemma-algebraically-closed (factorization over an algebraic closure)

Source reference. Original reference 09HP: sources/stacks-fields/pinned-original-source.tex, source lines 2068–2068. Prerequisite pointer invoked at 2333; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GU: sources/stacks-fields/pinned-original-source.tex, source lines 1240–1240. Prerequisite pointer invoked at 2330; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a finite extension of fields. There exists a unique smallest finite extension K/E such that K is normal over F.*

Proof. Choose generators $\alpha_1, \ldots, \alpha_n$ of E over F. Let $P_1, \ldots, P_n$ be the minimal polynomials of $\alpha_1, \ldots, \alpha_n$ over F. Set $P = P_1 \ldots P_n$. Observe that $(x-\alpha_1)\ldots(x-\alpha_n)$ divides P, since each $(x-\alpha_i)$ divides P_i. Say $P = (x-\alpha_1)\ldots(x-\alpha_n)Q$. Let K/E be the splitting field of P over E. We claim that K is the splitting field of P over F as well (which implies that K is normal over F). This is clear because K/E is generated by the roots of Q over E and E is generated by the roots of $(x-\alpha_1)\ldots(x-\alpha_n)$ over F, hence K is generated by the roots of P over F.

Uniqueness. Suppose that K'/E is a second smallest extension such that K'/F is normal. Choose an algebraic closure $\overline{F}$ and an embedding $\sigma_0 : E \to \overline{F}$. By

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 09GU we can extend σ_0 to $\sigma : K \to \overline{F}$ and $\sigma' : K' \to \overline{F}$. By Lemma 09HP we see that $\sigma(K) \cap \sigma'(K')$ is normal over F. By minimality we conclude that $\sigma(K) = \sigma'(K')$. Thus $\sigma^{-1} \circ \sigma' : K' \to K$ gives an isomorphism of extensions of E. $\square$